Semester Test 2

Electricity and Electronics EBN 111

9 June 2021

Test Information			
Maximum marks:	46	Full marks:	45
Duration of test:	100 min	Open/Closed book:	Open
Total number of pages (this page included)		17	

Lecturers: Dr D. le Roux, Dr S. Smith, Dr X. Ye

Assistant Lecturers: L. Kokot, M. Maritz, J. Masela, J. Thiselton, M. Trivedi, D. Wepener

Important

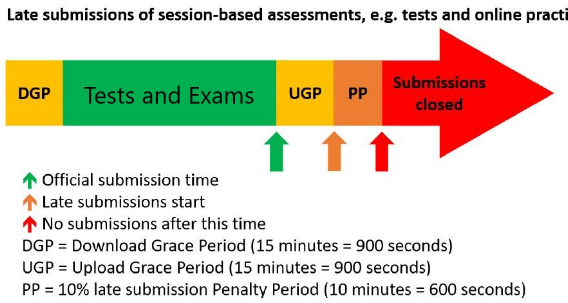
1. The examination regulations of the University of Pretoria apply.
2. Write your answers on the Excel spreadsheet found on the AMS system. The spreadsheet contains unique parameter values for each student. Therefore, you must download the Excel spreadsheet from your own AMS account. Follow the instructions on the AMS spreadsheet carefully.
3. Final answers must be written as real numbers and not as fractions. Use at least 5 significant figures to write numbers. Use decimal points and NOT decimal commas.
4. Answers must include units where applicable. (Refer to the EECE Rules for Electronic Assessments of tests and examinations for guidelines on how to enter units. Also refer to the example list of units on the Excel spreadsheet.)
5. **By uploading your semester test, you declare the following:**

   The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, examinations and/or any other forms of assessment. I am aware that no student or any other person may assist or attempt to assist another student, or obtain help, or attempt to obtain help from another student or any other person during tests, examinations and/or any other forms of assessment.

Online assessment submission rules and time-line

The following rules and time-line apply for all tests for EBN111/122 as illustrated by the figure below:

1. The Download Grace Period (DGP) is a 15 minute period before the official start of the test for students to download all documents related to the test and read all the necessary instructions.
2. The start of the green section after the DGP indicates the official start of the test. Students must work through the test at an average rate of at least 2 minutes per mark.
3. The upward green arrow at the end of the green section indicates the official submission deadline for the test.
4. Students must aim to submit their Excel spreadsheet with their answers on the AMS before the official submission deadline.
5. Students can request assistance via **e-mail** ebn2021.ams@gmail.com if they experience any technical issues with respect to uploading their Excel spreadsheets on the AMS.
6. An Upload Grace Period (UGP) of 15 minutes follows the official submission deadline. Students may submit their Excel spreadsheets with answers during this period without incurring any penalties. No email submissions will be accepted during the UGP.
7. A Penalty Period (PP) of 10 minutes follows the UGP. If a student submits an Excel spreadsheets during this period, the student incurs a 10% penalty to their final test mark.
8. No late submissions is accepted after the PP.



Additional guide for Complex numbers

The following examples illustrate how to appropriately enter complex numbers on the Excel Spreadsheet:

- **Correct**: *1-j2* or *-3+j4*
  - This is the rectangular form of a complex number. This format is acceptable as a single number entry in the numeric field.
  - Capital J is however incorrect, and *1-2j* is also not acceptable.
  - If a complex number has no real part, for example *0+j2*, write *0+j2* as the final answer.
- **Correct**:
  - With angle in degrees: 12∠34°
  - With angle in radians: 12∠0.5934
  - A complex number in its polar form is regarded as a single number entry with the angle in degrees or radians.
- **Incorrect**:
  - *1-i2* (Use j instead of i)
  - *1 - j2* (no spaces between characters)
  - 12 ∠ 34 ° (no spaces between characters)
  - *j10+1* (Real part before the imaginary part for rectangular form.)
  - *-3+4j* (should be $j4$ instead of $4j$)

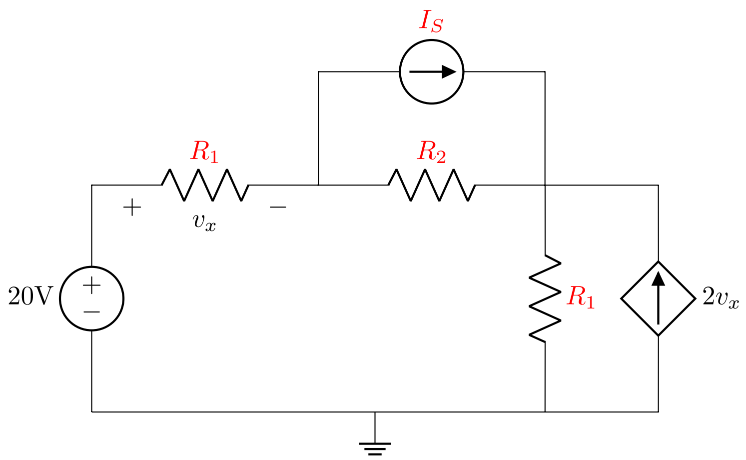
Section 1

Questions 1 to 3 [4]

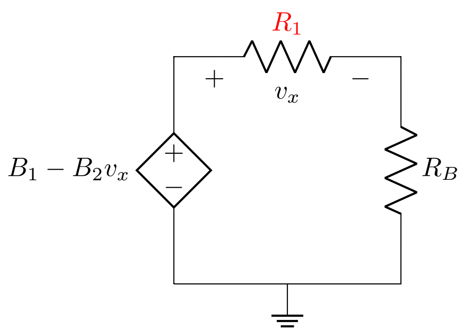
Given: I_S, R_1, R_2.

Use source transformation to obtain Circuit B so that it is equivalent to Circuit A, and determine R_B, B_1 and B_2 in Circuit B.

Circuit A:



Circuit B:



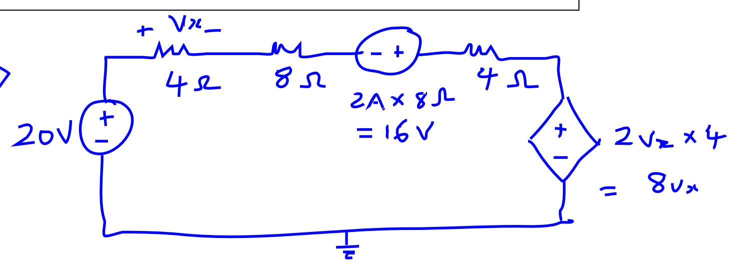
01	R_B [2]
02	B_1 [1]
03	B_2 [1]

* Given: $I_S = 2A$

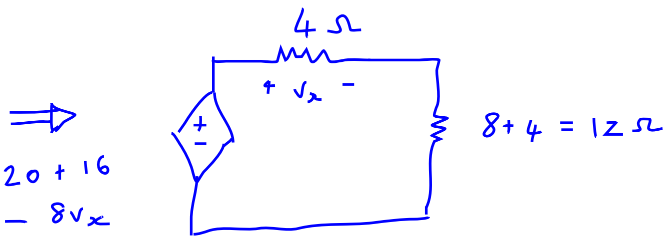
$R_1 = 4\ \Omega$

$R_2 = 8\ \Omega$

$\Rightarrow$



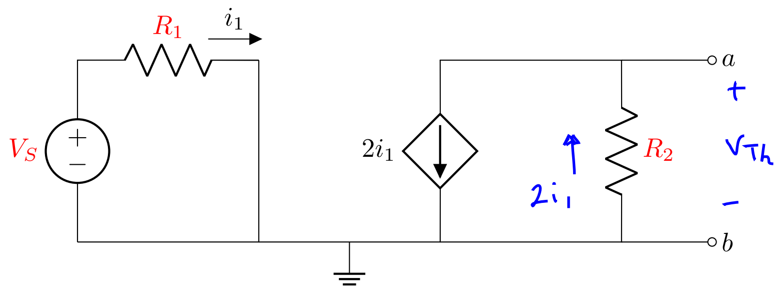
$\Rightarrow$



$R_B = 12\ \Omega$

$B_1 = 36$

$B_2 = 8$

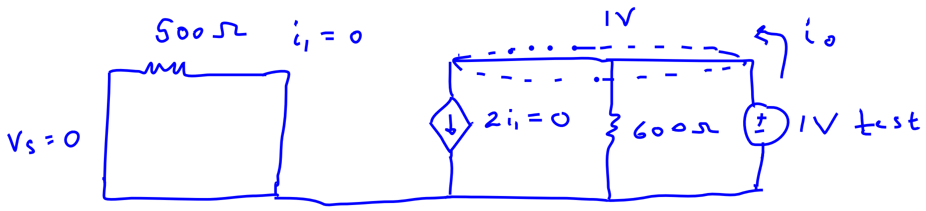
Questions 4 to 5 [4]	
Given: V_S, R_1, R_2.	
Determine the Thevenin equivalent voltage V_{Th} and resistance R_{Th} between terminals a-b.	
	
04	V_{Th} [2]
05	R_{Th} [2]

* Given: $V_S = 10\text{ V}$
$R_1 = 500\,\Omega$
$R_2 = 600\,\Omega$

V_{Th}: $i_1 = \dfrac{10\text{ V}}{500\,\Omega} = 20\text{ mA}$

$V_{Th} = -2i_1 \times 600\,\Omega$
$= -2(20\text{ mA}) \times 600\,\Omega$
$= -24\text{ V}$

R_{Th}:

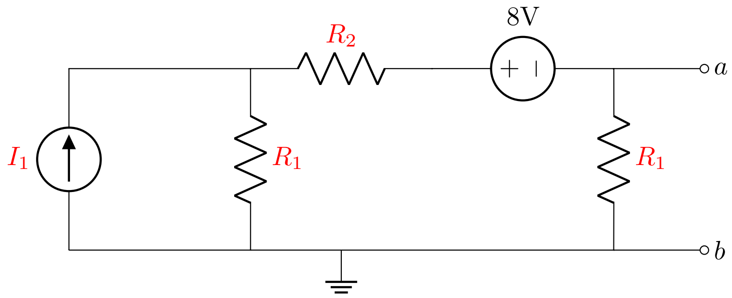


$R_{Th} = \dfrac{1\text{ V}}{i_0}$

KCL: $i_0 = \dfrac{1\text{ V}}{600\,\Omega} + \cancel{2i_0} \rightarrow 0$

$i_0 = \dfrac{1\text{ V}}{600\,\Omega}$

$\therefore R_{Th} = \dfrac{1\text{ V}}{\frac{1\text{ V}}{600\,\Omega}} = 600\,\Omega$

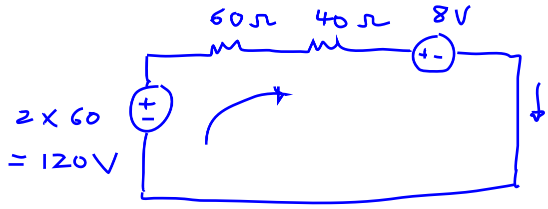
Questions 6 to 7 [4]	
Given: I_1, R_1, R_2.	
Determine the Norton equivalent current I_N and resistance R_N between terminals a-b.	
	
06	I_N [2]
07	R_N [2]

* Given: $I_1 = 2\,A$

$R_1 = 60\,\Omega$

$R_2 = 40\,\Omega$

I_N: source transf:

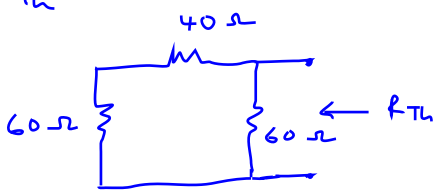


I_N (R_1 shorted)

KVL: $-120V + 100\,I_N + 8\,V = 0$

$$I_N = \frac{120-8}{100} = \frac{112}{100} = 1{,}12\,A$$

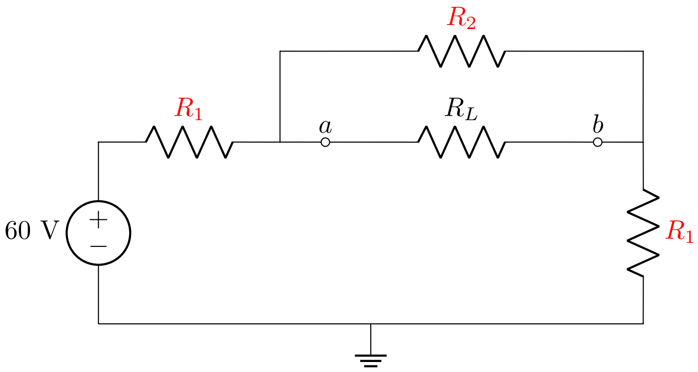
R_N: $R_N = R_{Th}$



$$R_{Th} = (60+40) \parallel 60$$

$$= 100 \parallel 60$$

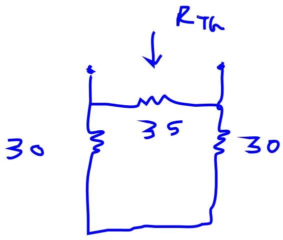
$$= \frac{6000}{160} = 37.5\,\Omega$$

Questions 8 to 9 [4]	
Given: R_1, R_2.	
Determine the load resistor R_L connected between terminals *a*-*b* for maximum power transfer.	
	
08	R_L [2]
Using the value of R_L calculated in Question 8, determine the maximum power P_{max}.	
09	P_{max} [2]

* Given: $R_1 = 30\,\Omega$

$R_2 = 35\,\Omega$

$R_L = R_{Th}$ for max power :

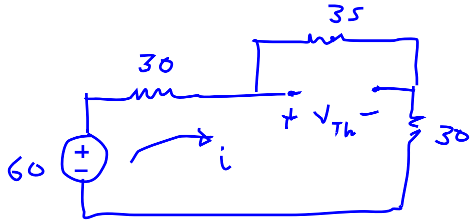


$R_{Th} = (30 + 30) \parallel 35 = 60 \parallel 35 = \frac{2100}{95}$

$R_{Th} = 22.105\,\Omega$

$P_{max} = \frac{V_{Th}^2}{4 R_{Th}}$

V_{Th} :



KVL : $-60 + 30i + 35i + 30i = 0$

$\therefore 95i = 60$

$i = \frac{60}{95} = 0.63158$ A

$V_{Th} = 35\,\Omega \times i$

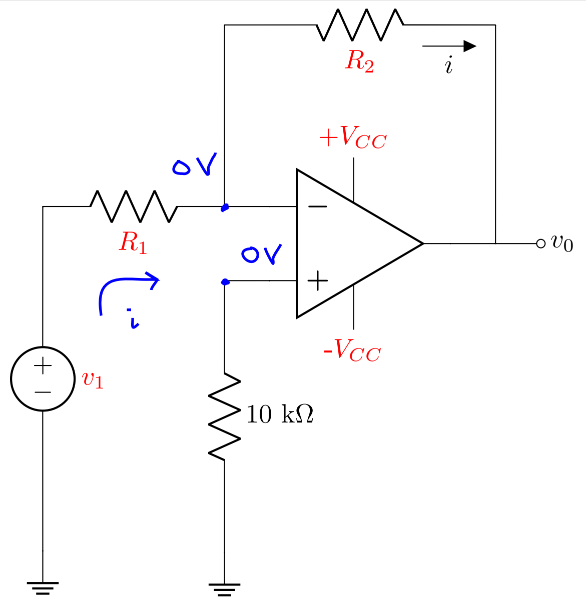
$= 35 \times 0.63158$ A $= 22.105$ V

$P_{max} = \frac{(22.105)^2}{4 \times 22.105}$

$P_{max} = 5.5263$ W

Total Section 1: [16]

Section 2

Questions 10 to 11 [4]	
Given: v_1, R_2.	
Assume that the op-amp is ideal.	
	
Given R_1, determine the current i.	
10	i [2]
Given V_{CC}, determine the value of R_1 just before the op-amp goes into negative saturation.	
11	R_1 [2]

* Given: $v_1 = 2V$
$R_2 = 5k\Omega$

Given $R_1 = 2k\Omega$: $i = \frac{2V}{2k\Omega}$ (no current flows into op-amp)

$i = 1mA$

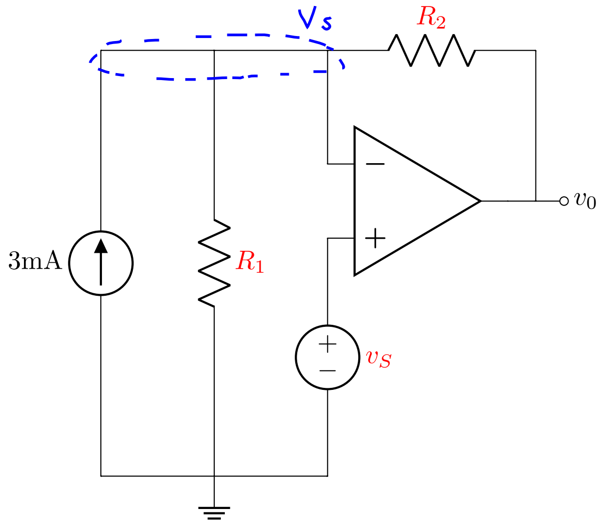
Given $V_{cc} = 16V$: $V_o = -2V\left(\frac{5k\Omega}{R_1}\right)$ (inverting configuration)

$\therefore V_o R_1 = -10\,000$

and $V_o = -V_{cc}$ for negative saturation

$\therefore R_1 = \frac{-10000}{-V_{cc}} = \frac{-10000}{-16}$ $\therefore R_1 = 625\ \Omega$

Question 12 [2]
Given: v_S, R_1, R_2.
Assume that the op-amp is ideal. Determine the voltage v_0.



12	v_0 [2]

* Given: $V_s = 10\ V$

$R_1 = 7\ k\Omega$

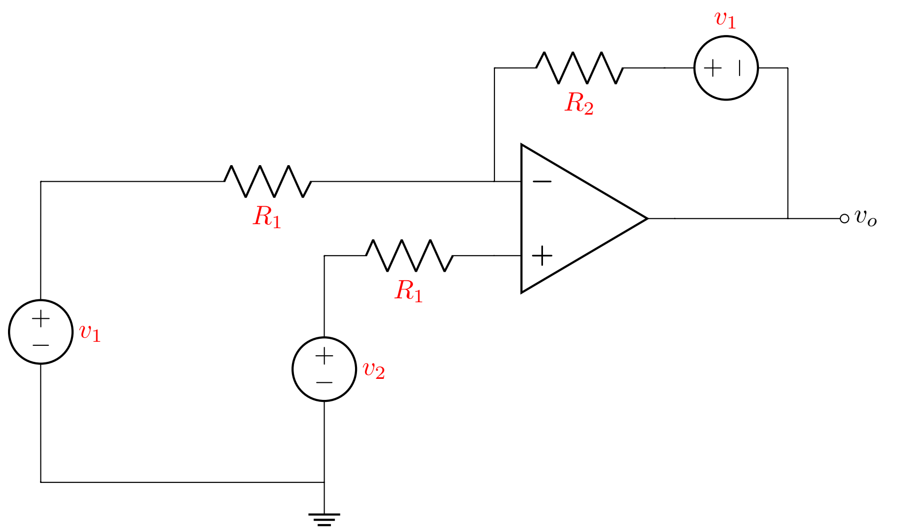
$R_2 = 8\ k\Omega$

KCL at node Vs : $-3mA + \frac{10V}{7k\Omega} + \frac{10V - V_0}{8k\Omega} = 0$

$(\times 56k): -168 + 80 + 70 - 7V_0 = 0$

$7V_0 = 150 - 168 = -18$

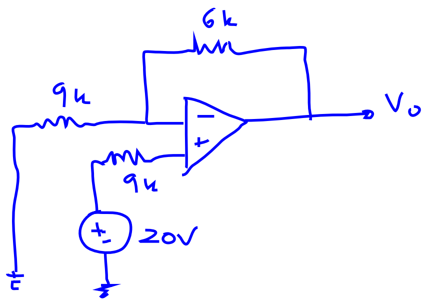
$V_0 = \frac{-18}{7} = -2.5714\ V$

Questions 13 to 14 [4]	
Given: R_1, R_2.	
Assume that the op-amp is ideal.	
	
Given: v_2 and $v_1 = 0$ V, determine the output voltage v_0.	
13	v_0 [2]
Given: v_1 and $v_2 = 0$ V, determine the output voltage v_0.	
14	v_0 [2]

* Given $R_1 = 9\,k\Omega$

$R_2 = 6\,k\Omega$

Given $V_2 = 20V$, $V_1 = 0V$:

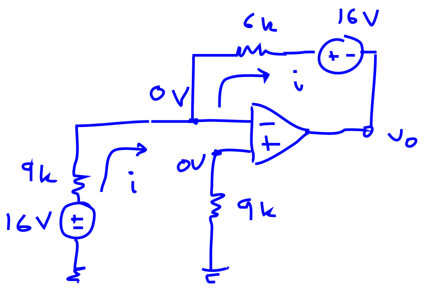


non-inverting: $V_o = 20\left(1 + \frac{6k}{9k}\right)$

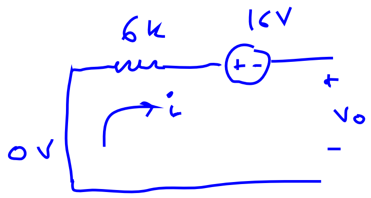
$V_o = 20(1.66667)$

$V_o = 33.333\ V$

Given $V_1 = 16V$, $V_2 = 0V$:



$i = \frac{16V}{9k} = 1.77778\ mA$



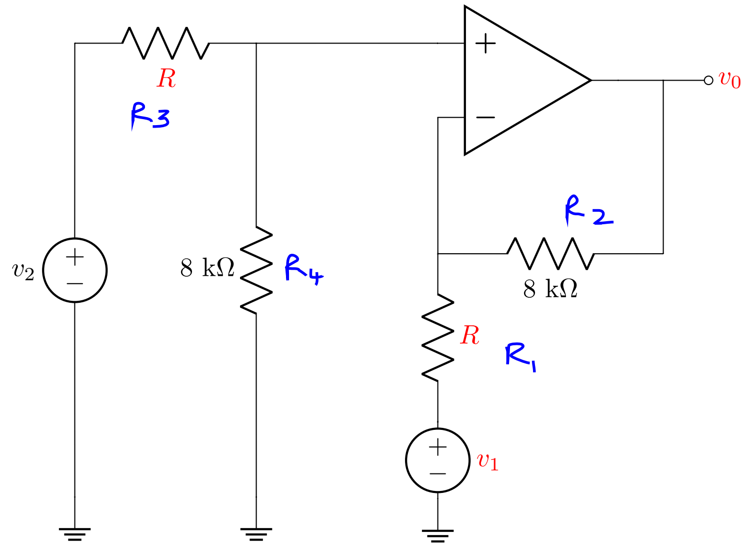
KVL: $6k(i) + 16 + V_o = 0$

$\therefore V_o = -10.667 - 16 = -26.6667\ V$

Question 15 [2]

Given: R, v_1, v_0.

Assume that the op-amp is ideal. Determine v_2.



15	v_2 [2]

✱ Given: $R = 2\,k\Omega$

$V_1 = 18\ V$

$V_o = 18\ V$

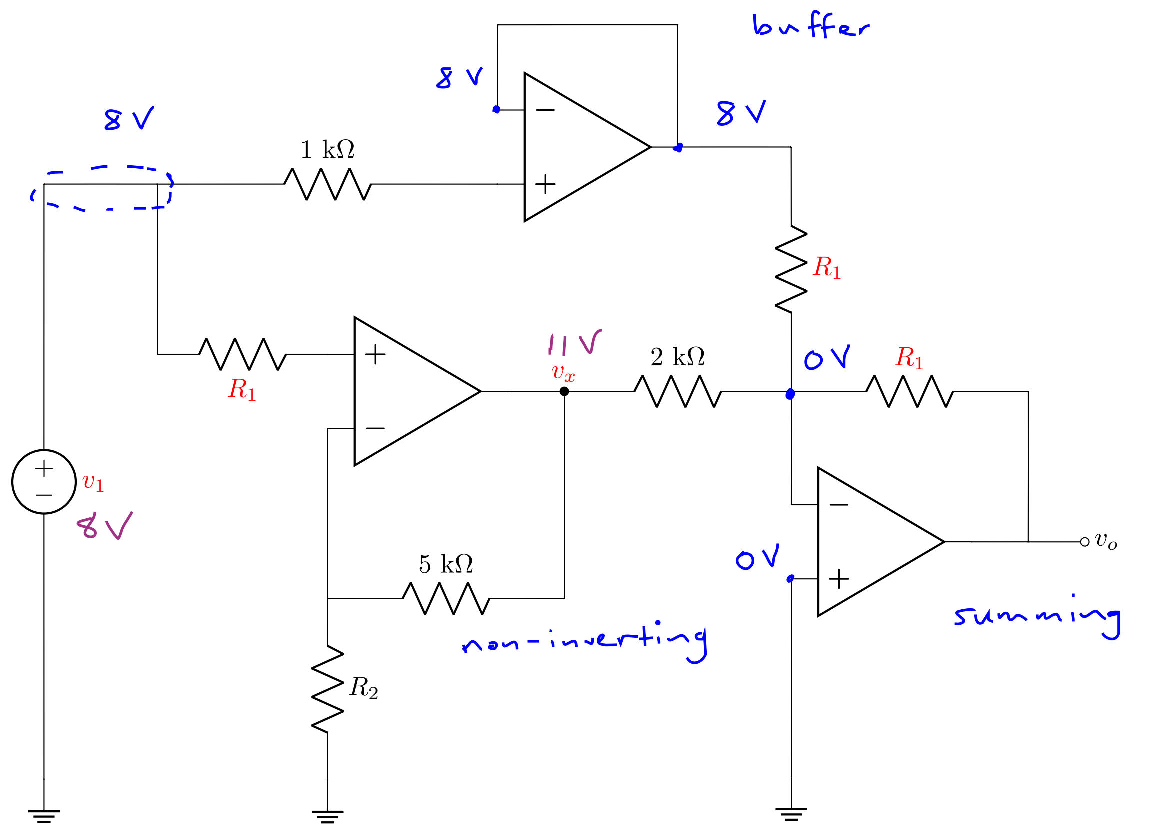
Difference amplifier: $\dfrac{R_1}{R_2} = \dfrac{R_3}{R_4} = \dfrac{2\,k\Omega}{8\,k\Omega}$

$$\therefore V_o = \frac{R_2}{R_1}(V_2 - V_1)$$

$$\therefore 18 = \frac{8\,k\Omega}{2\,k\Omega}(V_2 - 18)$$

$$18 = 4V_2 - 72$$

$$V_2 = \frac{90}{4} = \underline{22.5\,V}$$

Questions 16 to 17 [4]	
Given: v_1, R_1, v_x.	
Assume that the op-amps are ideal. Calculate the resistor R_2 and the voltage v_0.	
	
16	R_2 [2]
17	v_0 [2]

* Given: $V_1 = 8\text{ V}$
$R_1 = 6\text{ k}\Omega$
$V_x = 11\text{ V}$

$\underline{R_2}$: non-inverting

$$\therefore 11\text{ V} = 8\text{ V}\left(1 + \frac{5k}{R_2}\right)$$

$$\therefore 11 = 8 + \frac{40k}{R_2}$$

$$R_2 = \frac{40k}{11-8} = \frac{40k}{3} = 13.3333\text{ k}\Omega$$

$\underline{V_o}$: summing:

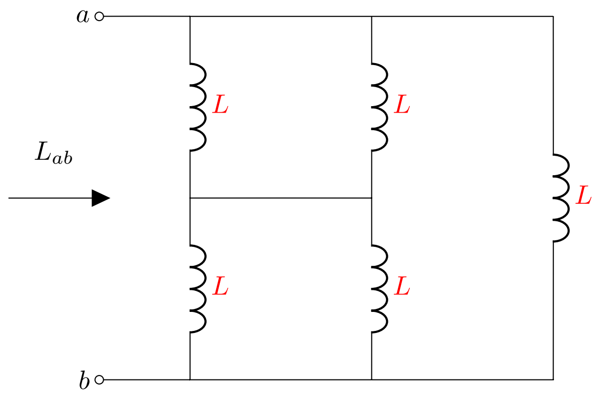
$$V_o = -\left(\frac{6k}{6k}(8\text{V}) + \frac{6k}{2k}(11\text{V})\right)$$

$$V_o = -(8+33) = -41\text{ V}$$

Total Section 2: [16]

Section 3

Question 18 [2]
Given: L
Determine the equivalent inductance L_{ab} between terminals a-b.



18	L_{ab} [2]

* Given: $L = 100\,mH$

$$L_{ab} = \left[\left(100\,mH \parallel 100\,mH\right) + \left(100\,mH \parallel 100\,mH\right)\right] \parallel 100\,mH$$

$$= \left(50\,mH + 50\,mH\right) \parallel 100\,mH$$

$$= 100\,mH \parallel 100\,mH$$

$$\therefore L_{ab} = 50\,mH$$

Question 19 [2]	
Given: C, A, t.	
The voltage across a capacitor C is equal to $v(t) = A\cos(1000t)$ V. Determine the current $i(t)$ through the capacitor at time t.	
19	$i(t)$ [2]

* Given: $C = 100\,\mu F$

$A = 6$

$t = 3\,s$

$$i = C\frac{dv}{dt}$$

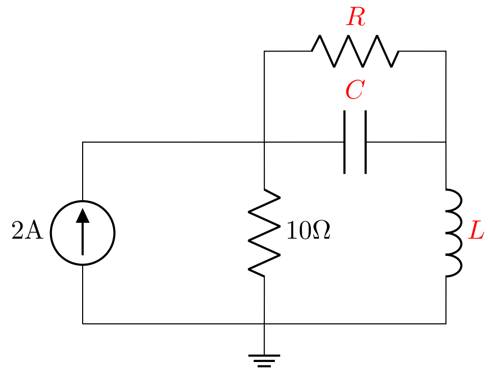
$$i(t) = (100 \times 10^{-6})\,6\,(-1000\sin(1000t))$$

$$i(3) = (600 \times 10^{-6})(-1000\sin(3000))$$

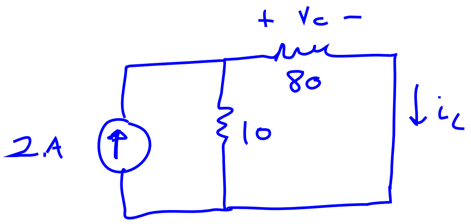
$$i(3) = -0.6\sin(3000)$$

$$i(3) = -0.6(0.21919)$$

$$i(3) = -0{,}131514\ A \quad \text{or} \quad -131.514\ mA$$

Questions 20 to 21 [4]	
Given: C, L, R.	
Under DC conditions, determine the following:	
	
The energy ω_L stored in the inductor L.	
20	ω_L [2]
The energy ω_C stored in the capacitor C.	
21	ω_C [2]

* Given: $C = 300\,\mu F$
$L = 10\,mH$
$R = 80\,\Omega$



$i_L = 2\left(\frac{10}{10+80}\right)$ current division

$i_L = \frac{20}{90} = 0.22222\ A$

$V_c = 80\, i_L = 17.777\ V$

$\omega_L = \frac{1}{2} L\, i_L^2 = 5mH\,(0.2222)^2 = 246.908\ \mu J$

$\omega_c = \frac{1}{2} C\, V_c^2 = 150\mu F\,(17.7776)^2 = 47.4065\ mJ$

Total Section 3: [8]

Section 4

Question 22 [2]	
Given: R.	
Determine $\mathbf{V}_2$ and write the answer in ***rectangular form*** ($\mathbf{V}_2 = x + jy$). $$\begin{bmatrix} R+j2 & -jR \\ -jR & R \end{bmatrix} \begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} = \begin{bmatrix} 0 \\ R\angle 90^\circ \end{bmatrix}$$	
22	$\mathbf{V}_2$ in ***rectangular form***. [2]

* Given $R = 4$

$$\begin{bmatrix} 4+j2 & -j4 \\ -j4 & 4 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 0 \\ j4 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 4+j2 & -j4 \\ -j4 & 4 \end{vmatrix} = 4(4+j2) - (-j4)(-j4)$$

$$= 16 + j8 + 16 = 32 + j8$$

$$\Delta_2 = \begin{vmatrix} 4+j2 & 0 \\ -j4 & j4 \end{vmatrix} = j4(4+j2) = j16 - 8$$

$$V_2 = \frac{\Delta_2}{\Delta} = \frac{-8+j16}{32+j8}$$

$$V_2 = -0.11764 + j0.52941$$

Question 23 [2]	
Given: X, Y.	
Simplify the following expression for $\mathbf{Z}$ and write the answer in ***polar form*** ($\mathbf{Z} = r\angle\theta^{\circ}$): $$\mathbf{Z} = \frac{\sqrt{-jX}}{(\sqrt{Y}+j)^2(-X+jY)}$$ Give the answer as $r \geq 0$ and the angle in the range -180° $\leq \theta \leq$ +180°.	
23	$\mathbf{Z}$ in ***polar form***. [2]

\# Given: $X = 4$

$Y = 2$

$$Z = \frac{\sqrt{-j4}}{(\sqrt{2}+j)^2(-4+j2)} = \frac{2\angle -45^{\circ}}{(1+j2\sqrt{2})(-4+j2)}$$

$$Z = \frac{2\angle -45^{\circ}}{-4+j2-j8\sqrt{2}-4\sqrt{2}} = \frac{2\angle -45^{\circ}}{-4-4\sqrt{2}+j(2-8\sqrt{2})}$$

$$Z = 0.14907\angle 91.0362^{\circ}$$

Questions 24 to 25 [2]	
Given: A, B.	
Three sinusoids are given as follows: $$v_1(t) = A\sin\left(200t + B^{\circ}\right)\text{ V}$$ $$v_2(t) = 5\cos\left(200t + B^{\circ}\right)\text{ V}$$ $$v_3(t) = -A\sin\left(200t + B^{\circ}\right)\text{ V}$$	
Consider $v_1(t)$ and $v_2(t)$ and determine which lags the other. Answer = 1 if $v_1(t)$ lags $v_2(t)$. Answer = 2 if $v_2(t)$ lags $v_1(t)$.	
24	Write 1 or 2 in the answer block [1]
Consider $v_2(t)$ and $v_3(t)$ and determine which leads the other. Answer = 1 if $v_2(t)$ leads $v_3(t)$. Answer = 2 if $v_3(t)$ leads $v_2(t)$.	
25	Write 1 or 2 in the answer block [1]

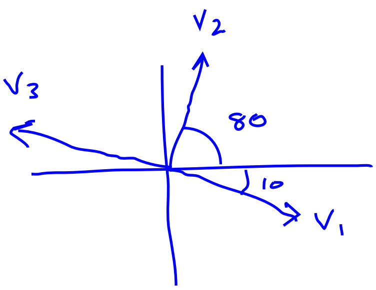
* Given: $A = 4$

$B = 80$

$$V_1 = 4\cos\left(80^{\circ} - 90^{\circ}\right) = 4\cos\left(-10^{\circ}\right)$$

$$V_2 = 5\cos\left(80^{\circ}\right)$$

$$V_3 = 4\cos\left(80^{\circ} + 90^{\circ}\right) = 4\cos\left(170^{\circ}\right)$$



24 Answer = 1 $\longrightarrow$ (V_1 lags V_2)

25 Answer = 2 $\longrightarrow$ (V_3 leads V_2)

Total Section 4: [6]

Grand Total : [46]